When Machine Learning Beats Persistence: A Leakage-Free Streamflow and Drought Benchmark for the Tropical Andes

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Abstract Reported machine-learning gains in Andean streamflow forecasting are often inflated by evaluation leakage. We publish an open benchmark of 19 long-record gauges from the Andean Glacierized Catchment dataset: 7 on the Altiplano, including Puente Isla Cabanillas in the Lake Titicaca basin, 6 in the Cordillera Blanca and 6 in the western and southern Andes. The gauges are combined with ERA5 and MERRA-2 forcing and GloFAS v4 simulated discharge. Persistence and climatology are compared with six learners at 1-, 3- and 7-day leads. The protocol is calendar-continuous, embargoed and validation-tuned, with block-bootstrap confidence intervals, Diebold–Mariano tests and walk-forward refits. The learners beat persistence at 15 of 19 gauges at one day and at 18 of 19 at seven days (median skill 0.101). At Cabanillas the one-day skill is 0.110 (95% confidence interval 0.064–0.144), the opposite of what an earlier leaky pipeline reported. Yet a lower root-mean-square error does not improve the detection of drought days: the median critical success index is 0.43 for persistence and 0.30 for the learners. Hydrological drought decorrelates over a few hundred kilometres, and skill follows hydroclimatic regime rather than geography, so dense gauge networks matter more than distant proxies.

Key words: Tropical Andes · Altiplano · Streamflow forecasting · Hydrological drought · Machine learning · Data leakage · Benchmark

1 Introduction

Streamflow in the tropical Andes sustains irrigation, hydropower and drinking water for millions of people, yet it is shaped by an extreme seasonal cycle, retreating glaciers and recurrent drought [3, 23, 55]. On the Altiplano, the endorheic plateau shared by Peru, Bolivia and Chile, rivers such as the Coata–Cabanillas feed Lake Titicaca and depend almost entirely on a short austral-summer wet season [11];

climate projections point to more frequent and longer hydrological droughts in the Titicaca–Desaguadero–Poopó system [58]. Short-range discharge forecasts and early drought signals are therefore valuable. In practice, however, gauges are sparse, records are gappy and operational tools are rarely open.

Machine learning (ML) is increasingly used for data-driven streamflow forecasting [20,21,31,37,56]. For daily lead times, however, discharge is strongly autocorrelated, and the naive persistence forecast $\hat{Q}_{t+h} = Q_t$ is a demanding benchmark [35, 38]. Reported gains can be inflated by (i) comparing models against weak references, (ii) randomly shuffled cross-validation that leaks temporal information [4, 40, 54], and (iii) implementation details that give baselines and learners different information at issue time. An earlier version of our own pipeline suffered from the third problem: the learners saw discharge only up to $t-1$ while persistence used Q_t, and calendar gaps were treated as consecutive days. Correcting both reversed the one-day conclusion, and this experience motivated the explicit protocol reported here.

This paper contributes an open, laptop-scale benchmark for the Peruvian Andes and the Altiplano in which every evaluation choice is explicit and tested. Specifically, we:

1. assemble 19 long-record gauges from the Andean-GC dataset [2], covering three regions: 7 Altiplano gauges (Cabanillas, Peru, and the Lauca/Caquena headwaters, Chile), 6 glacier-fed Cordillera Blanca gauges and 6 western/southern Andean gauges. We combine them with forcing from ERA5, the fifth-generation reanalysis of the European Centre for Medium-Range Weather Forecasts (ECMWF) [17]; with precipitation from NASA's Prediction of Worldwide Energy Resources (POWER) service, which is based on the Modern-Era Retrospective analysis for Research and Applications, version 2 (MERRA-2); and with discharge simulated by the Global Flood Awareness System (GloFAS) v4;
2. define a leakage-free protocol: a calendar-continuous index, identical information at issue time for all methods, chronological splits with an embargo, hyperparameter selection on validation only, and expanding-window walk-forward checks;
3. report skill against persistence with moving-block bootstrap intervals [22] and Diebold–Mariano tests [7, 16], at 1-, 3- and 7-day leads;
4. evaluate drought detection directly, using the variable-threshold level method [9, 49] and categorical scores [41], and release a nonparametric standardized streamflow index (SSI) [8].

The paper is organized as follows. Section 2 reviews related work, Sect. 3 describes the data and Sect. 4 the methods. Section 5 presents the results, Sect. 6 discusses them and Sect. 7 concludes.

2 Related Work

Data-driven streamflow forecasting in the Andes. Random forests have been used for runoff and flash-flood forecasting in Ecuadorian Andean catchments [33], recently combined with satellite precipitation and hydrological knowledge [32]. Support-vector regression was tested for monthly runoff in the Central Andes of Argentina [46]. In Peru, artificial neural networks have forecast monthly flows in the high-Andean Crisnejas basin [51], and k-nearest neighbors daily flows of the Ramis River on the Altiplano [26]. Ensemble data assimilation with a conceptual model improved hourly forecasts in the Vilcanota basin [24]. On the Altiplano, Lujano et al. found that reanalysis precipitation performs well against gauges, but gridded forcing can still degrade conceptual-model streamflow [25]. This result motivates both our reanalysis-based features and the data-fusion ablation. Large-sample studies with long short-term memory (LSTM) networks set the methodological reference for data-driven hydrology [1, 12, 20, 21, 37]. Most Andean studies, however, evaluate a single basin at a single lead time, using NSE without a persistence reference or an uncertainty estimate. Our prior multi-station climate-ML study of Altiplano frost [48] used the same comparative-benchmark design.

Validation under dependence. Random k-fold cross-validation is optimistic for autocorrelated or spatially structured data [4, 40]. For Peruvian agricultural survey data, blocked cross-validation revealed that random folds overstated accuracy [54], and hierarchical spatial models were needed to characterize climate vulnerability [27]. We transfer the same principle to time: chronological blocks with an embargo, and walk-forward refits.

Hydrological drought. Hydrological drought is a below-normal availability of streamflow or storage [29, 49]. It is identified with threshold-level methods [9, 50] or with standardized runoff and streamflow indices (SRI, SSI) [42, 53]. These are analogues of the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI) [28, 52]. Parametric SSI values are sensitive to the chosen distribution [47], which motivates nonparametric plotting-position formulations [8, 13]. Drought forecasting requires an explicit identification step after the streamflow forecast, and the chosen definition changes the result [15, 43, 44]. We therefore score drought days explicitly instead of inferring them from RMSE.

3 Study Area and Data

Discharge. We use quality-controlled daily discharge from the Andean-GC dataset [2], released under CC BY 4.0. We keep every Peruvian gauge, operated by the National Meteorological and Hydrological Service (SENAMHI), and every Altiplano gauge of the Lauca/Caquena headwaters in Chile, operated by the General Directorate of Water (DGA), with at least 5000 quality-controlled (QC) days, which gives 19 gauges (13 Peru, 6 Chile; Table 1, Fig. 1). The focus gauge is Puente

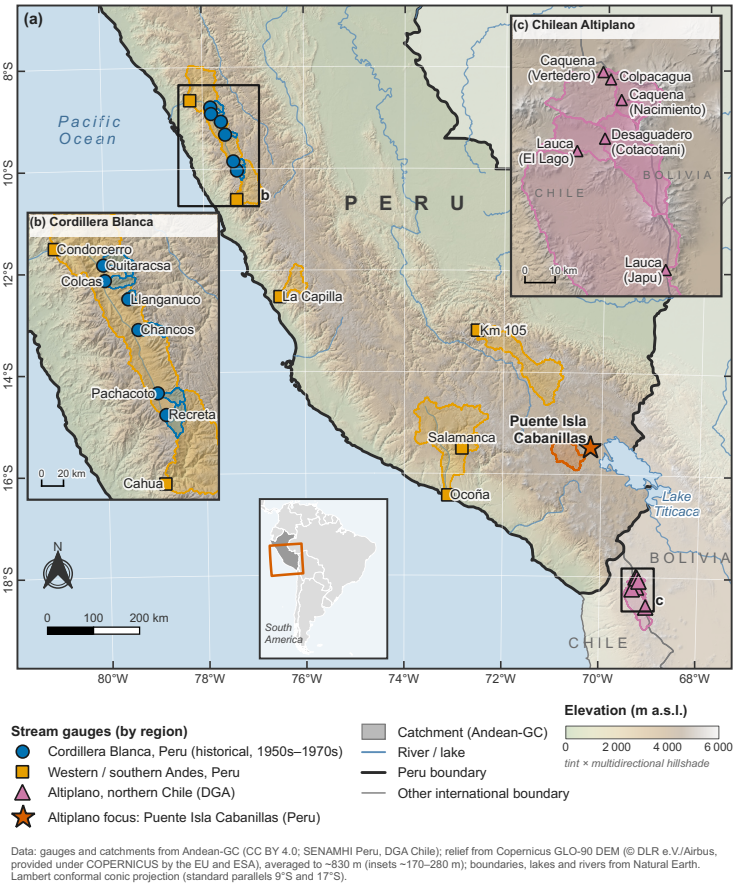


Fig. 1 The 19 benchmark gauges and their Andean-GC catchments, with marker shape and shading indicating the region. The Altiplano focus gauge Puente Isla Cabanillas is highlighted. Insets show the Cordillera Blanca and Chilean Altiplano clusters.

Isla Cabanillas (`P00000055`; 2843 km^2, mean elevation 4471 m), which drains the Coata–Cabanillas system into Lake Titicaca. The Cordillera Blanca gauges are historical records from 1952–1974 in strongly glacierized catchments. The remaining gauges span 1950–2024. Records contain multi-year gaps (e.g., 1996–2005 at Cabanillas), which we keep as missing values on a daily calendar.

Meteorological forcing. Daily precipitation and 2-m temperature come from ERA5 [17] at each catchment centroid, retrieved through the Open-Meteo historical API [57]. The API's ERA5-Land endpoint [34] does not provide precipitation, so we use ERA5 throughout. A second, independent precipitation estimate is NASA POWER `PRECTOTCORR` (MERRA-2 bias-corrected with GPCP; available from 1981).

Simulated discharge. GloFAS v4 reanalysis discharge is retrieved from the Open-Meteo Flood API (available from 1984) on a 3×3 neighborhood of 0.05° cells around each gauge. The cell used is the one with the highest Kling–Gupta efficiency (KGE) over the fitting period only, so the test period is never used for this choice.

Table 1 Benchmark gauges (Andean-GC, $\geq$5000 QC days). Area in km^2; n_{test} is the number of test days at a 7-day lead. ⋆Focus gauge.

ID	Gauge	Cty	Region	Area	Record	QC days	n_{test}
P00000055	Puente Isla Cabanillas⋆	Pe	Altiplano	2,843	1994–2024	6,572	941
X01001001	Caquena @ source	Ch	Altiplano	50	1976–2004	7,350	997
X01001002	Caquena @ weir	Ch	Altiplano	468	1969–2024	15,812	2007
X01001003	Colpacagua @ mouth	Ch	Altiplano	199	1988–2024	10,181	1407
X01020002	Desaguadero Cotacotani	Ch	Altiplano	376	1964–2017	16,165	2028
X01020003	Lauca @ El Lago	Ch	Altiplano	583	1953–2024	14,474	2019
X01021001	Lauca @ Japu	Ch	Altiplano	3,278	1963–2022	13,732	1719
P00000003	Chancos	Pe	Cordillera Blanca	218	1953–1974	6,739	836
P00000005	Colcas	Pe	Cordillera Blanca	239	1954–1974	6,796	909
P00000011	Llanganuco	Pe	Cordillera Blanca	82	1955–1974	6,398	826
P00000016	Pachacoto	Pe	Cordillera Blanca	209	1952–1974	6,701	868
P00000021	Quitaracsa	Pe	Cordillera Blanca	388	1952–1974	6,928	943
P00000022	Recreta	Pe	Cordillera Blanca	346	1953–1974	7,481	1038
P00000027	Cahua	Pe	W/S Andes	3,388	2000–2024	9,131	1357
P00000008	Condorcerro	Pe	W/S Andes	10,618	1973–2024	17,517	2595
P00000047	Egemsa Km 105	Pe	W/S Andes	9,601	1985–2024	14,308	2124
P00000031	La Capilla	Pe	W/S Andes	2,153	1950–2024	18,420	1880
P00000040	Ocoña	Pe	W/S Andes	16,017	1970–2024	5,701	755
P00000041	Salamanca	Pe	W/S Andes	1,108	1965–2018	15,352	1455

4 Methods

Forecast set-up. For each gauge and lead $h \in \{1,3,7\}$ days, a forecast issued at day t predicts Q_{t+h} using information available at t. The 27 predictors are: discharge lags Q_{t-k}, $k \in \{0,1,2,3,7,14,30\}$; the one-day change; rolling means and standard deviations over 7, 14 and 30 days ending at t; precipitation lags (0–7 days) and 7-, 30- and 90-day totals; 2-m temperature (daily, 7- and 30-day means); and the day of year encoded as sine and cosine. All operations use a daily calendar index, so a lag of k days is exactly k days. Any sample whose predictors or target fall in a data gap is dropped. Persistence uses the same Q_t that the learners see.

Methods compared. There are two references: persistence and a day-of-year climatology (15-day smoothed mean of the fitting period, evaluated at the target date). We compare six learners: Ridge regression [18], random forest [5], histogram gradient boosting [10], XGBoost [6], a two-layer multilayer perceptron (MLP), and a low-flow-oriented XGBoost trained on the change in $\log(1+Q)$ (XGBoost-log). For every learner, a small grid of hyper-parameters is searched together with the target form: the level Q_{t+h} or the change $Q_{t+h} - Q_t$.

Validation protocol. Each record is split chronologically into 70% training, 15% validation and 15% test. The last h samples of the training and validation blocks

are removed (embargo) so that no target overlaps the next block. Configurations are selected on validation RMSE. The selected configuration is refit on training plus validation and scored once on the test block. When we refer to the learner of a gauge, we mean the one with the lowest validation RMSE, so the test set is never used to choose models. As a robustness check, an expanding-window walk-forward with three folds is run for Ridge at every gauge, re-selecting hyper-parameters within each fold.

Scores and uncertainty. We report the root-mean-square error (RMSE), the Nash–Sutcliffe efficiency (NSE) [36], the KGE [14, 19] and the RMSE skill against persistence, $S = 1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{pers}}$ [35]. 95% confidence intervals (CIs) for S come from a paired moving-block bootstrap (1000 replicates, 30-day blocks) that preserves autocorrelation [22, 39]. Equal squared-error accuracy against persistence is tested with the Diebold–Mariano test using the small-sample correction [7, 16].

Drought detection. Following the variable-threshold level method [9, 44, 49], a day is in hydrological drought when Q falls below the monthly 20th percentile (Q80 exceedance) of the fitting period. For each method, we compare the forecast drought state at $t + h$ with the observed state and compute the probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI) [41]. For description we also compute a nonparametric SSI-30: the 30-day mean discharge is ranked within each calendar month, converted to a Gringorten plotting position, and mapped to a standard-normal quantile [8, 13]. This avoids the distributional sensitivity of parametric SSI [47].

Data-fusion ablation. Where MERRA-2 precipitation and GloFAS overlap the record (1984 onwards), we refit Ridge and XGBoost on identical samples with two predictor sets: ERA5 only, and ERA5 plus MERRA-2 precipitation and lagged GloFAS discharge up to t. GloFAS values at the target date are never used as predictors. We also report the raw GloFAS simulation as a physically based reference.

Spatial statistics. Spatial drought coherence is quantified as the Spearman correlation of monthly SSI-30 between every gauge pair with at least 60 overlapping months, as a function of great-circle distance d. We fit the exponential correlogram $\rho(d) = c\,e^{-d/d_0}$, in the spirit of space–time drought analysis [45]. Pairs share gauges and are therefore not independent, so the 95% interval of the decorrelation distance d_0 comes from a gauge-level bootstrap (500 replicates). Spatial clustering of the 7-day skill is tested with global Moran's I [30], using inverse-distance weights and 9999 permutations. Candidate drivers of skill are Andean-GC catchment attributes (area, mean elevation, glacier fraction, aridity, precipitation concentration, snow fraction and dry-day frequency). They are screened with Spearman correlations, permutation p-values and a Holm correction for multiple testing. The skill map and the drought-synchrony network are rendered in QGIS.

The whole pipeline runs on a laptop CPU (10 cores, about 4 min), and every number in this paper is written automatically from the result tables, so text and results cannot drift apart.

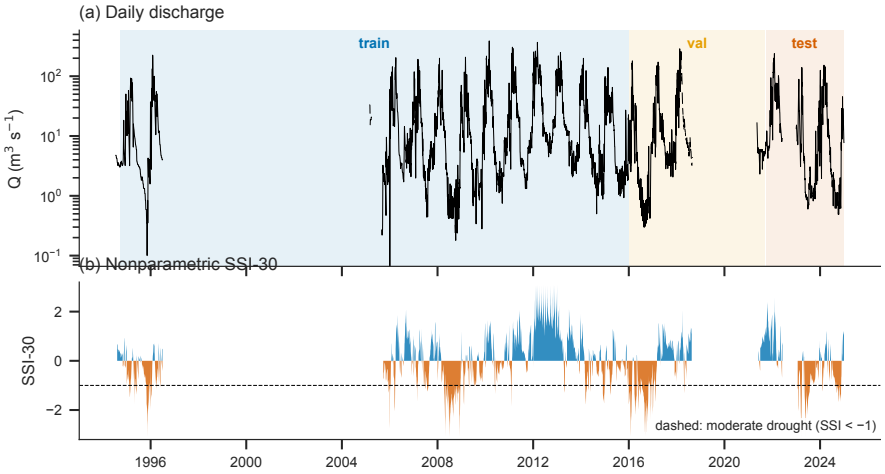


Fig. 2 Puente Isla Cabanillas: daily discharge (log scale) with the chronological train/validation/test blocks, and the nonparametric SSI-30 (bars below zero: below-normal flow). Gaps are real missing records.

5 Results

5.1 Focus gauge: Puente Isla Cabanillas

Figure 2 shows the Cabanillas record, the chronological blocks and the nonparametric SSI-30. Table 2 reports all methods on the test period (2021-09-22 to 2024-12-24). With identical information at issue time, persistence is no longer unbeatable at one day. The validation-selected random forest reaches NSE = 0.922 against 0.902 for persistence, a skill of 0.110 (95% CI 0.064 to 0.144; Diebold–Mariano $p < 0.001$). This reverses the one-day conclusion of the earlier, leaky version of the pipeline (Sect. 1). At seven days, persistence drops to NSE = 0.461. Ridge regression on the change target gives the most robust gain (skill 0.115, CI excluding zero), and its walk-forward folds give skills of 0.166, 0.114, 0.114. The learner selected on validation at this lead (XGBoost-log) has a smaller, non-significant skill of 0.038. Validation-based selection is unbiased, but it is noisy when validation and test hydrology differ.

5.2 Multi-station benchmark

Across the 19 gauges, the validation-selected learner beats persistence at 15/19 gauges at one day, 15/19 at three days and 18/19 at seven days. Its 95% interval excludes zero at 9, 10 and 12 gauges, respectively (Fig. 3). The median skill grows with lead time, from 0.046 to 0.080 and 0.101, because persistence degrades faster

Table 2 Test-period scores at Puente Isla Cabanillas. Skill is relative to persistence with 95% block-bootstrap CI. DM: Diebold–Mariano p-value. CSI: drought-day detection. †Learner selected on validation RMSE.

h	Method	RMSE	NSE	KGE	Skill	95% CI	DM p	CSI
1	Persistence	11.30	0.902	0.951	0.000	–	–	0.88
	Climatology	28.29	0.383	0.535	−1.504	[−2.12, −0.82]	<0.001	0.00
	Ridge	10.38	0.917	0.944	0.081	[0.03, 0.12]	0.003	0.65
	RandomForest†	10.05	0.922	0.931	0.110	[0.06, 0.14]	<0.001	0.87
	HistGB	10.46	0.916	0.933	0.074	[0.01, 0.14]	0.089	0.58
	XGBoost	10.49	0.915	0.899	0.071	[−0.01, 0.13]	0.088	0.84
	MLP	9.71	0.927	0.941	0.141	[0.09, 0.18]	<0.001	0.39
	XGBoost-log	10.43	0.916	0.950	0.077	[0.00, 0.13]	0.052	0.89
3	Persistence	19.98	0.693	0.846	0.000	–	–	0.73
	Climatology	28.25	0.386	0.536	−0.414	[−0.72, −0.07]	0.006	0.00
	Ridge	17.99	0.751	0.829	0.100	[0.04, 0.15]	0.040	0.58
	RandomForest	18.29	0.742	0.813	0.084	[−0.01, 0.15]	0.146	0.67
	HistGB	18.61	0.733	0.825	0.069	[−0.08, 0.18]	0.317	0.72
	XGBoost	19.55	0.706	0.774	0.021	[−0.12, 0.14]	0.751	0.64
	MLP	18.75	0.729	0.838	0.061	[0.03, 0.09]	0.028	0.68
	XGBoost-log†	18.49	0.737	0.823	0.074	[−0.03, 0.15]	0.228	0.77
7	Persistence	26.46	0.461	0.731	0.000	–	–	0.61
	Climatology	28.17	0.389	0.538	−0.065	[−0.32, 0.21]	0.682	0.00
	Ridge	23.41	0.578	0.677	0.115	[0.03, 0.21]	0.112	0.47
	RandomForest	24.78	0.528	0.686	0.064	[−0.02, 0.14]	0.389	0.59
	HistGB	27.05	0.437	0.619	−0.022	[−0.16, 0.12]	0.812	0.18
	XGBoost	27.76	0.407	0.628	−0.049	[−0.19, 0.09]	0.600	0.13
	MLP	32.18	0.203	0.536	−0.216	[−0.48, −0.01]	0.098	0.36
	XGBoost-log†	25.45	0.502	0.581	0.038	[−0.08, 0.18]	0.652	0.56

than the learners (Fig. 4). The regions behave differently. Skill rises steeply with lead time in the Cordillera Blanca (median 0.032 to 0.118) and in the western/southern Andes (0.047 to 0.110). On the Altiplano it stays flat (0.081 to 0.057), and the smallest Chilean headwater (Caquena at its source, 50 km^2) remains negative at seven days. Model choice matters less than the protocol: Ridge alone is skillful at 18/19 gauges at seven days. The change target was selected in 68% of the learner fits. The expanding-window walk-forward confirms the pattern: 53 of 57 gauge–fold combinations have positive seven-day Ridge skill (median 0.103).

5.3 Hydrological-drought detection

Lower RMSE does not translate into better drought detection (Fig. 5). At seven days, the median CSI of drought days is 0.43 for persistence and 0.30 for the validation-selected learner. The learner beats persistence at only 2/19 gauges. The low-flow-oriented XGBoost-log does not close the gap (median CSI 0.33; better than persistence at 3 gauges). The learners are more conservative: they raise fewer false alarms (median FAR 0.35 vs. 0.39) but miss more drought days (POD 0.42 vs. 0.60). The reason is that RMSE-trained models regress towards the conditional mean and smooth the recession limb, near where the threshold lies. This agrees with the view that drought forecasting needs its own identification and verification step [44].

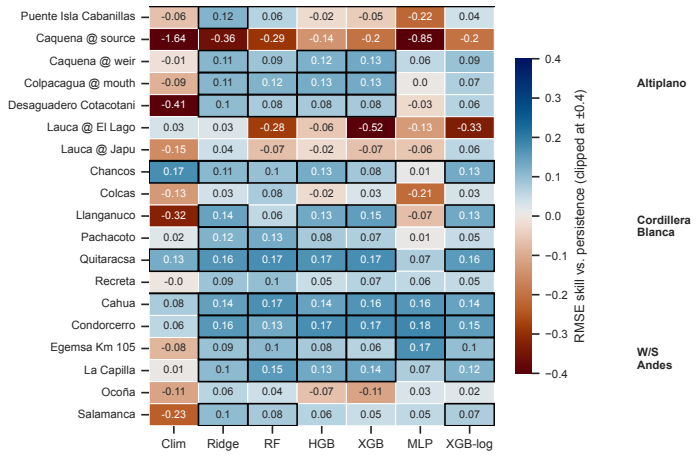


Fig. 3 Seven-day RMSE skill against persistence for every gauge and method. Boxed cells: 95% moving-block bootstrap interval above zero. Gauges are grouped by region (top to bottom: Altiplano, Cordillera Blanca, western/southern Andes).

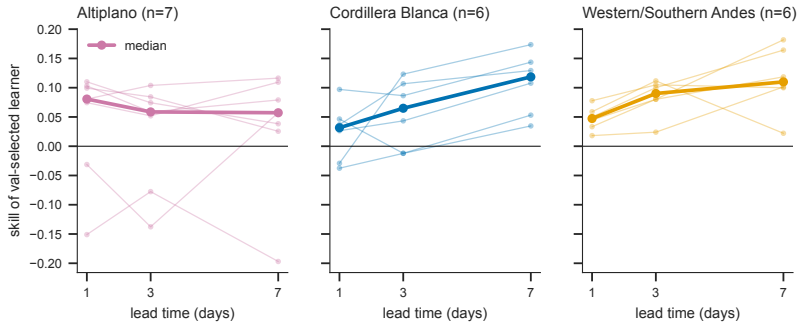


Fig. 4 Skill of the validation-selected learner versus lead time, by region (thin lines: gauges; thick: median).

5.4 Data fusion and the GloFAS reference

At the 7 gauges where MERRA-2 and GloFAS overlap the record, adding them to the ERA5 predictors changes the seven-day skill by a median of -0.004 for Ridge and -0.008 for XGBoost. Only 2 and 2 gauges improve, respectively. The raw GloFAS v4 simulation, even with the cell chosen on the fitting period, has a median NSE of -0.10 (positive at 1/7 gauges). At these small, steep Andean catchments, recent observed discharge carries far more short-range information than a global model or a second reanalysis.

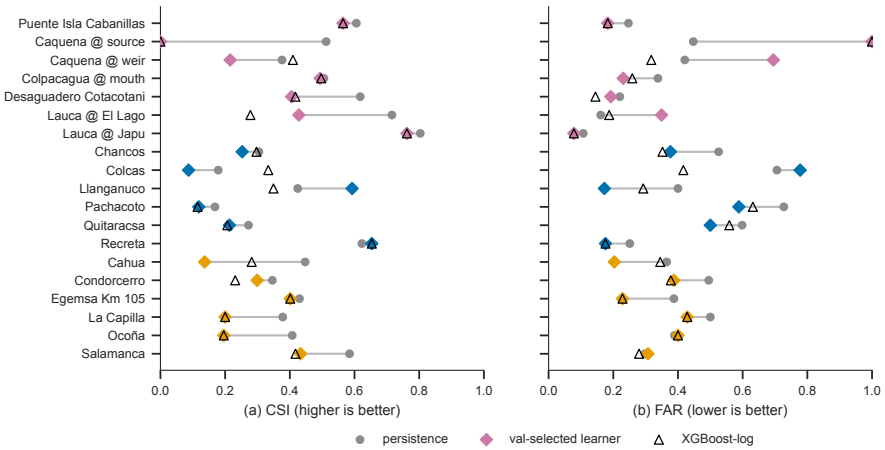


Fig. 5 Seven-day detection of drought days (variable threshold: monthly Q20 of the fitting period). Left: CSI; right: false-alarm ratio.

5.5 Spatial statistics

Hydrological drought is spatially coherent but local (Figs. 6b and 7a). Across 104 gauge pairs, the monthly SSI-30 correlation decreases with distance (Spearman -0.56, $p < 0.001$). It follows $\rho(d) = 0.35\, e^{-d/d_0}$ with a decorrelation distance $d_0 =$ 582 km (95% CI 330–1069 km). Pairs within the same region correlate more strongly than pairs across regions (median ρ 0.28 vs. 0.09). Even nearby gauges share only part of their drought signal, which shows that one gauge cannot represent drought on the Altiplano or in the Cordillera Blanca. Forecast skill, in contrast, shows no spatial clustering: Moran's $I = 0.018$ ($p = 0.374$; expectation under randomness -0.056; Fig. 6a). Skill is better explained by hydroclimatic regime than by location (Fig. 7b). The strongest correlate is snow fraction (ρ =-0.57, permutation $p = 0.010$, Holm-adjusted $p = 0.071$), followed by precipitation concentration and aridity. Skill tends to be lower in catchments with more snow, more concentrated precipitation and higher aridity, which is the hydroclimate of the Altiplano headwaters. However, only snow fraction approaches significance after correction for multiple testing.

6 Discussion

Protocol before models. The largest change in our conclusions came from the protocol, not from model choice. Giving persistence and learners the same information at issue time, and indexing on the calendar, turned a reported one-day loss into a significant gain. Hyper-parameter selection on validation and bootstrap intervals then separated robust gains from noise. We recommend that Andean ML forecasting stud-

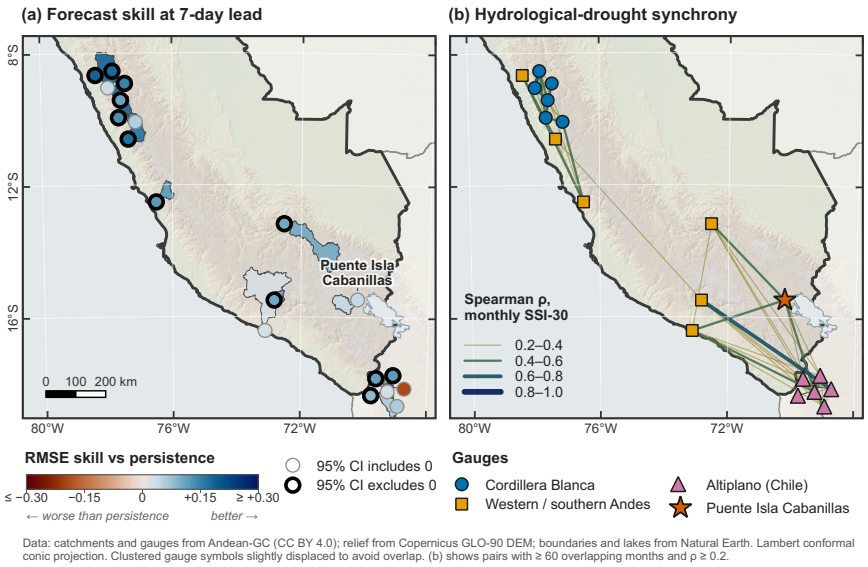


Fig. 6 QGIS maps of (a) seven-day skill of the validation-selected learner by catchment (bold outline: 95% CI above zero) and (b) the hydrological-drought synchrony network (monthly SSI-30 Spearman correlation between gauge pairs with $\geq$60 overlapping months).

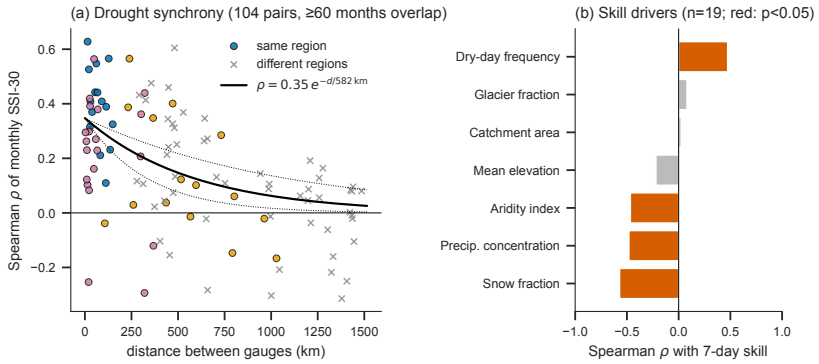


Fig. 7 (a) Drought-synchrony correlogram with exponential fit (dotted: 95% gauge-bootstrap range of d_0). (b) Spearman correlation between seven-day skill and catchment attributes (highlighted bars: permutation $p < 0.05$, uncorrected).

ies report skill against persistence with uncertainty, publish their split and embargo design, and test leakage explicitly; our test suite does this automatically.

Where ML helps. Gains are modest, a median RMSE reduction of about 5–10%, but consistent: positive at almost every gauge at seven days and confirmed by walk-forward folds. Linear models on the change target are competitive with tree ensembles, in line with large-sample findings that careful baselines are hard to beat at

short leads [37,38]. The Altiplano is the hardest region. Its headwaters combine short convective wet-season pulses with long baseflow recessions [11]. A 0.25° reanalysis can represent this rainfall only partially, and even well-performing gridded products degrade conceptual streamflow simulations there [25].

Drought is a different target. Better RMSE did not yield better drought-day detection. Operational drought early warning should train or calibrate on the drought decision itself, for example with quantile or classification losses and threshold-aware post-processing, and verify it with categorical scores [15,44]. The limited spatial coherence of drought (d_0 of a few hundred kilometres) argues for dense gauge networks rather than extrapolating from single stations, consistent with projected changes in the Titicaca system [58].

Limitations. ERA5 at 0.25° is coarse for steep catchments and has a wet bias over the Andes. The Cordillera Blanca records predate 1975, so no MERRA-2 or GloFAS data overlap them. The fusion ablation therefore covers 7 gauges only, and some GloFAS requests were rate-limited. The validation block is a single period, so selection can be unstable under non-stationarity (e.g., glacier retreat). Forecasts use observed rather than forecast forcing beyond t, which is acceptable here because predictors stop at t; skill against operational meteorological forecasts is left for future work.

7 Conclusion

We released an open, leakage-free benchmark for short-range streamflow and hydrological drought forecasting at 19 Andean gauges with an Altiplano focus. Under an honest protocol, data-driven models beat persistence at most gauges and leads: at seven days at 18/19 gauges (median skill 0.101), and also at one day at the Altiplano focus gauge, Cabanillas. RMSE gains do not carry over to drought-day detection, where persistence remains a strong reference. Hydrological drought decorrelates over about 582 km, and forecast skill depends on hydroclimatic regime rather than location. Future work will add drought-aware training losses, operational meteorological forecasts as inputs, and more Altiplano gauges from SENAMHI and the National Water Authority (ANA).

Data and code availability

The discharge data are from Andean-GC (doi:10.5281/zenodo.18035801; CC BY 4.0). ERA5 and GloFAS were retrieved through Open-Meteo [57], and MERRA-2 precipitation through NASA POWER. Source code, data extracts, leakage tests, QGIS map scripts, result tables and supplementary figures are public at:
`https://github.com/Andre031222/streamflow-drought-altiplano`

Declarations

Author contributions. R. A. Vilca-Solorzano conceived the study, developed the software and the analysis, and wrote the manuscript. R. Quispe-Vargas validated the results and revised the manuscript. D. M. Yana-Yucra supervised the study, contributed to the methodology and analysis, and revised the manuscript. All authors approved the final manuscript.

Competing interests. The authors declare no competing interests.

Ethics and intended use. No human subjects or personal data were involved. The forecasts are research benchmarks and do not replace the official products of SENAMHI, DGA or other national services.

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References

1. Addor, N., Newman, A.J., Mizukami, N., Clark, M.P.: The CAMELS data set: catchment attributes and meteorology for large-sample studies. Hydrology and Earth System Sciences **21**(10), 5293–5313 (2017). DOI 10.5194/hess-21-5293-2017
2. Aguayo, R., Zekollari, H., van Tiel, M., Bolibar, J., Van Tricht, L., Ayala Ramos, A.I., Ultee, L.: Andean Glacierized Catchment (Andean-GC) dataset. Zenodo dataset (2026). DOI 10.5281/zenodo.18035801
3. Baraer, M., Mark, B.G., McKenzie, J.M., Condom, T., Bury, J., Huh, K.I., Portocarrero, C., Gómez, J., Rathay, S.: Glacier recession and water resources in Peru's Cordillera Blanca. Journal of Glaciology **58**(207), 134–150 (2012). DOI 10.3189/2012JoG11J186
4. Bergmeir, C., Benítez, J.M.: On the use of cross-validation for time series predictor evaluation. Information Sciences **191**, 192–213 (2012). DOI 10.1016/j.ins.2011.12.028
5. Breiman, L.: Random forests. Machine Learning **45**(1), 5–32 (2001). DOI 10.1023/A:1010933404324
6. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. ACM, New York (2016). DOI 10.1145/2939672.2939785
7. Diebold, F.X., Mariano, R.S.: Comparing predictive accuracy. Journal of Business & Economic Statistics **13**(3), 253–263 (1995). DOI 10.1080/07350015.1995.10524599
8. Farahmand, A., AghaKouchak, A.: A generalized framework for deriving nonparametric standardized drought indicators. Advances in Water Resources **76**, 140–145 (2015). DOI 10.1016/j.advwatres.2014.11.012
9. Fleig, A.K., Tallaksen, L.M., Hisdal, H., Demuth, S.: A global evaluation of streamflow drought characteristics. Hydrology and Earth System Sciences **10**(4), 535–552 (2006). DOI 10.5194/hess-10-535-2006
10. Friedman, J.H.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics **29**(5), 1189–1232 (2001). DOI 10.1214/aos/1013203451
11. Garreaud, R., Vuille, M., Clement, A.C.: The climate of the Altiplano: observed current conditions and mechanisms of past changes. Palaeogeography, Palaeoclimatology, Palaeoecology **194**(1–3), 5–22 (2003). DOI 10.1016/S0031-0182(03)00269-4

12. Gauch, M., Kratzert, F., Klotz, D., Nearing, G., Lin, J., Hochreiter, S.: Rainfall–runoff prediction at multiple timescales with a single Long Short-Term Memory network. Hydrology and Earth System Sciences **25**(4), 2045–2062 (2021). DOI 10.5194/hess-25-2045-2021
13. Gringorten, I.I.: A plotting rule for extreme probability paper. Journal of Geophysical Research **68**(3), 813–814 (1963). DOI 10.1029/JZ068i003p00813
14. Gupta, H.V., Kling, H., Yilmaz, K.K., Martinez, G.F.: Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. Journal of Hydrology **377**(1–2), 80–91 (2009). DOI 10.1016/j.jhydrol.2009.08.003
15. Hao, Z., Singh, V.P., Xia, Y.: Seasonal drought prediction: Advances, challenges, and future prospects. Reviews of Geophysics **56**(1), 108–141 (2018). DOI 10.1002/2016RG000549
16. Harvey, D., Leybourne, S., Newbold, P.: Testing the equality of prediction mean squared errors. International Journal of Forecasting **13**(2), 281–291 (1997). DOI 10.1016/S0169-2070(96)00719-4
17. Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R.J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., de Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.N.: The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society **146**(730), 1999–2049 (2020). DOI 10.1002/qj.3803
18. Hoerl, A.E., Kennard, R.W.: Ridge regression: Biased estimation for nonorthogonal problems. Technometrics **12**(1), 55–67 (1970). DOI 10.1080/00401706.1970.10488634
19. Knoben, W.J.M., Freer, J.E., Woods, R.A.: Technical note: Inherent benchmark or not? comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. Hydrology and Earth System Sciences **23**(10), 4323–4331 (2019). DOI 10.5194/hess-23-4323-2019
20. Kratzert, F., Klotz, D., Brenner, C., Schulz, K., Herrnegger, M.: Rainfall–runoff modelling using Long Short-Term Memory (LSTM) networks. Hydrology and Earth System Sciences **22**(11), 6005–6022 (2018). DOI 10.5194/hess-22-6005-2018
21. Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S., Nearing, G.: Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. Hydrology and Earth System Sciences **23**(12), 5089–5110 (2019). DOI 10.5194/hess-23-5089-2019
22. Künsch, H.R.: The jackknife and the bootstrap for general stationary observations. The Annals of Statistics **17**(3), 1217–1241 (1989). DOI 10.1214/aos/1176347265
23. Lavado Casimiro, W.S., Ronchail, J., Labat, D., Espinoza, J.C., Guyot, J.L.: Basin-scale analysis of rainfall and runoff in Peru (1969–2004): Pacific, Titicaca and Amazonas drainages. Hydrological Sciences Journal **57**(4), 625–642 (2012). DOI 10.1080/02626667.2012.672985
24. Llauca, H., Arestegui, M., Lavado-Casimiro, W.: Constraining flood forecasting uncertainties through streamflow data assimilation in the tropical Andes of Peru: Case of the Vilcanota river basin. Water **15**(22), 3944 (2023). DOI 10.3390/w15223944
25. Lujano, E., Diaz, R.D., Lujano, R., Sanchez-Delgado, M., Lujano, A.: Hydrological performance of gridded meteorological products in Peruvian Altiplano basins. RBRH – Revista Brasileira de Recursos Hídricos **30**, e10 (2025). DOI 10.1590/2318-0331.302520240068
26. Lujano, E., Lujano, R., Huamani, J.C., Lujano, A.: Hydrological modeling based on the KNN algorithm: An application for the forecast of daily flows of the Ramis river, Peru. Tecnología y Ciencias del Agua **14**(2), 169–203 (2023). DOI 10.24850/j-tyca-14-02-05
27. Maquera Andrade, A.J., Torres Cruz, F.: Bayesian spatial analysis of agricultural yields and climate vulnerability in southern Peru using hierarchical CAR models, 2024. In: S. Duarte Correia, M. Lahby (eds.) Green Technologies and Sustainable Intelligence—Volume II: Applications, Case Studies, and Future Challenges, *Springer Proceedings in Physics*, vol. 352, pp. 43–55. Springer Nature Switzerland, Cham (2026). DOI 10.1007/978-3-032-20151-5_4
28. McKee, T.B., Doesken, N.J., Kleist, J.: The relationship of drought frequency and duration to time scales. In: Proceedings of the 8th Conference on Applied Climatology, pp. 179–184. American Meteorological Society, Anaheim, CA (1993). No DOI assigned

29. Mishra, A.K., Singh, V.P.: A review of drought concepts. Journal of Hydrology **391**(1–2), 202–216 (2010). DOI 10.1016/j.jhydrol.2010.07.012
30. Moran, P.A.P.: Notes on continuous stochastic phenomena. Biometrika **37**(1–2), 17–23 (1950). DOI 10.1093/biomet/37.1-2.17
31. Mosavi, A., Ozturk, P., Chau, K.w.: Flood prediction using machine learning models: Literature review. Water **10**(11), 1536 (2018). DOI 10.3390/w10111536
32. Muñoz, P., Muñoz, D.F., Orellana-Alvear, J., Célleri, R.: Enhancing runoff forecasting through the integration of satellite precipitation data and hydrological knowledge into machine learning models. Natural Hazards **121**(4), 3915–3937 (2025). DOI 10.1007/s11069-024-06939-w
33. Muñoz, P., Orellana-Alvear, J., Willems, P., Célleri, R.: Flash-flood forecasting in an Andean mountain catchment—development of a step-wise methodology based on the random forest algorithm. Water **10**(11), 1519 (2018). DOI 10.3390/w10111519
34. Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N.: ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. Earth System Science Data **13**(9), 4349–4383 (2021). DOI 10.5194/essd-13-4349-2021
35. Murphy, A.H.: Skill scores based on the mean square error and their relationships to the correlation coefficient. Monthly Weather Review **116**(12), 2417–2424 (1988). DOI 10.1175/1520-0493(1988)116<2417:SSBOTM>2.0.CO;2
36. Nash, J.E., Sutcliffe, J.V.: River flow forecasting through conceptual models part I — a discussion of principles. Journal of Hydrology **10**(3), 282–290 (1970). DOI 10.1016/0022-1694(70)90255-6
37. Nearing, G.S., Kratzert, F., Sampson, A.K., Pelissier, C.S., Klotz, D., Frame, J.M., Prieto, C., Gupta, H.V.: What role does hydrological science play in the age of machine learning? Water Resources Research **57**(3), e2020WR028,091 (2021). DOI 10.1029/2020WR028091
38. Pappenberger, F., Ramos, M.H., Cloke, H.L., Wetterhall, F., Alfieri, L., Bogner, K., Mueller, A., Salamon, P.: How do I know if my forecasts are better? Using benchmarks in hydrological ensemble prediction. Journal of Hydrology **522**, 697–713 (2015). DOI 10.1016/j.jhydrol.2015.01.024
39. Politis, D.N., Romano, J.P.: The stationary bootstrap. Journal of the American Statistical Association **89**(428), 1303–1313 (1994). DOI 10.1080/01621459.1994.10476870
40. Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F.: Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography **40**(8), 913–929 (2017). DOI 10.1111/ecog.02881
41. Schaefer, J.T.: The critical success index as an indicator of warning skill. Weather and Forecasting **5**(4), 570–575 (1990). DOI 10.1175/1520-0434(1990)005<0570:TCSIAA>2.0.CO;2
42. Shukla, S., Wood, A.W.: Use of a standardized runoff index for characterizing hydrologic drought. Geophysical Research Letters **35**(2), L02,405 (2008). DOI 10.1029/2007GL032487
43. Sucozhañay, A., Timbe, L., Boll, J., Célleri, R.: Streamflow drought identification and characterization in a tropical Andean basin: effect of threshold methods. Hydrological Sciences Journal **71**(3), 391–409 (2026). DOI 10.1080/02626667.2025.2600086
44. Sutanto, S.J., Van Lanen, H.A.J.: Streamflow drought: implication of drought definitions and its application for drought forecasting. Hydrology and Earth System Sciences **25**(7), 3991–4023 (2021). DOI 10.5194/hess-25-3991-2021
45. Tallaksen, L.M., Hisdal, H., Van Lanen, H.A.J.: Space–time modelling of catchment scale drought characteristics. Journal of Hydrology **375**(3–4), 363–372 (2009). DOI 10.1016/j.jhydrol.2009.06.032
46. Teverovsky Korsic, S.A., Notarnicola, C., Uriburu Quirno, M., Cara, L.: Assessing a data-driven approach for monthly runoff prediction in a mountain basin of the Central Andes of Argentina. Environmental Challenges **10**, 100,680 (2023). DOI 10.1016/j.envc.2023.100680

47. Tijdeman, E., Stahl, K., Tallaksen, L.M.: Drought characteristics derived based on the Standardized Streamflow Index: A large sample comparison for parametric and nonparametric methods. Water Resources Research **56**(10), e2019WR026,315 (2020). DOI 10.1029/2019WR026315
48. Torres-Cruz, F., Yana-Yucra, D.M., Vilca-Solorzano, R.A.: Comparative analysis of statistical, machine learning, and deep learning approaches for frost prediction in the Peruvian Altiplano. International Journal of Advanced Computer Science and Applications **16**(9) (2025). DOI 10.14569/IJACSA.2025.0160992
49. Van Loon, A.F.: Hydrological drought explained. WIREs Water **2**(4), 359–392 (2015). DOI 10.1002/wat2.1085
50. Van Loon, A.F., Laaha, G.: Hydrological drought severity explained by climate and catchment characteristics. Journal of Hydrology **526**, 3–14 (2015). DOI 10.1016/j.jhydrol.2014.10.059
51. Vásquez-Ramírez, L., Vásquez-Paredes, L.F.: Predicción de caudales mensuales en ríos de cuencas altoandinas con enfoque de redes neuronales artificiales. Caso: río Crisnejas, Perú. Tecnología y Ciencias del Agua **14**(1), 124–199 (2023). DOI 10.24850/j-tyca-14-01-04
52. Vicente-Serrano, S.M., Beguería, S., López-Moreno, J.I.: A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index (SPEI). Journal of Climate **23**(7), 1696–1718 (2010). DOI 10.1175/2009JCLI2909.1
53. Vicente-Serrano, S.M., López-Moreno, J.I., Beguería, S., Lorenzo-Lacruz, J., Azorin-Molina, C., Morán-Tejeda, E.: Accurate computation of a streamflow drought index. Journal of Hydrologic Engineering **17**(2), 318–332 (2012). DOI 10.1061/(ASCE)HE.1943-5584.0000433
54. Vilca Apaza, W., Torres Cruz, F., Carpio Vargas, E.E.: Blocked and hierarchical cross-validation for agricultural data with spatial autocorrelation: Application to Peru's national agricultural survey. In: S. Duarte Correia, M. Lahby (eds.) Green Technologies and Sustainable Intelligence—Volume II: Applications, Case Studies, and Future Challenges, *Springer Proceedings in Physics*, vol. 352, pp. 3–13. Springer Nature Switzerland, Cham (2026). DOI 10.1007/978-3-032-20151-5_1
55. Vuille, M., Carey, M., Huggel, C., Buytaert, W., Rabatel, A., Jacobsen, D., Soruco, A., Villacis, M., Yarleque, C., Elison Timm, O., Condom, T., Salzmann, N., Sicart, J.E.: Rapid decline of snow and ice in the tropical Andes – impacts, uncertainties and challenges ahead. Earth-Science Reviews **176**, 195–213 (2018). DOI 10.1016/j.earscirev.2017.09.019
56. Yaseen, Z.M., El-shafie, A., Jaafar, O., Afan, H.A., Sayl, K.N.: Artificial intelligence based models for stream-flow forecasting: 2000–2015. Journal of Hydrology **530**, 829–844 (2015). DOI 10.1016/j.jhydrol.2015.10.038
57. Zippenfenig, P.: Open-Meteo.com Weather API. Zenodo software, version 1.4.0 (2024). DOI 10.5281/zenodo.7970649
58. Zubieta, R., Molina-Carpio, J., Laqui, W., Sulca, J., Ilbay, M.: Comparative analysis of climate change impacts on meteorological, hydrological, and agricultural droughts in the Lake Titicaca basin. Water **13**(2), 175 (2021). DOI 10.3390/w13020175